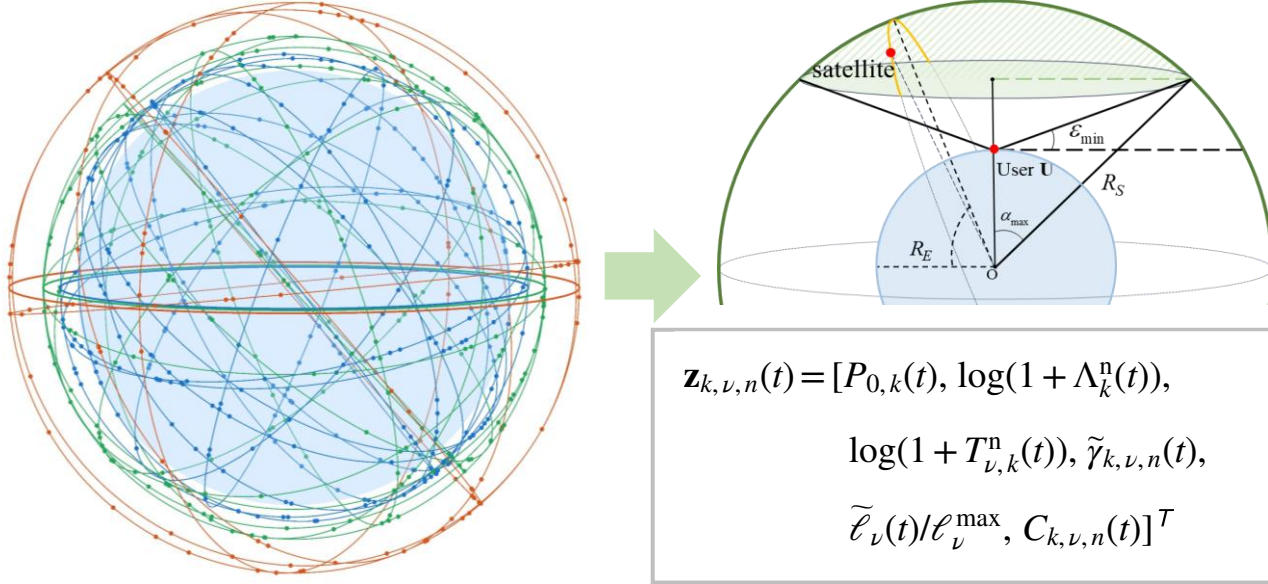


Physics-conditioned Temporal Graph Learning Framework

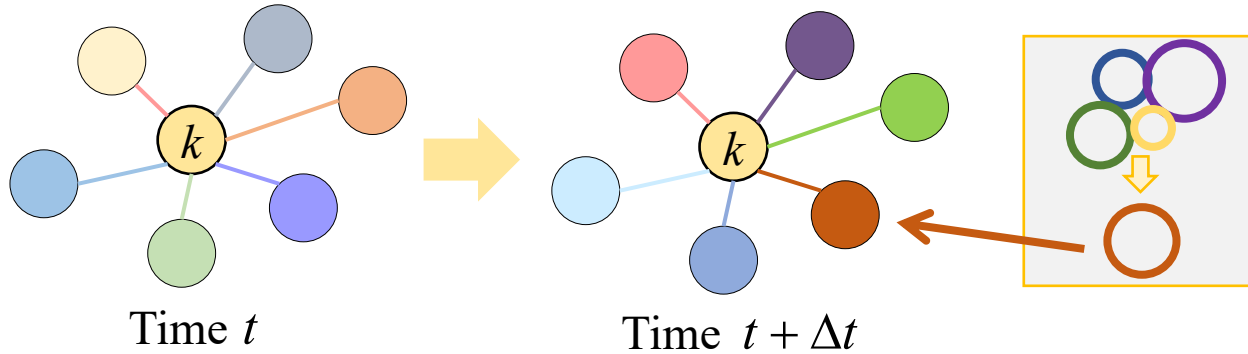
a. Physical model creation & descriptors



Cox-process modeling

P0: No-new-feasible-entry probability

c. Temporal Graph Network (TGN)



b. Physics-Conditioned Kernel Bank

Kernel Library $\mathbf{b}_m(\cdot)$

$\bigcirc \kappa_{\text{surv}}(\mathbf{z}) = \exp\left[-\rho\left(1 - e^{-\beta p_{\text{feas},k}(t) T_{v,k}^{\text{rst}}(t)}\right)\right]$
 $\bigcirc$ Trilinear 8

$\bigcirc$ Trilinear 1

$\bigcirc \kappa_{\text{load}}(\mathbf{z}) \triangleq \tilde{\ell}_v(t)/\ell_v^{\max}$

$\alpha_{k,v,n}(t) =$

$\text{softmax}\left(g_{\theta}^{\text{KAN}}\left([\mathbf{h}_k(t - \Delta t) \parallel \mathbf{h}_{v,n}(t - \Delta t) \parallel \mathbf{z}_{k,v,n}(t)]\right)\right)$

Coefficient Head

$\bigcirc$ surv
 $\bigcirc$ prod. 1
 $\bigcirc$ load
 $\bigcirc$ prod. 8
 $\dots$

Message Builder

$$\mathbf{m}_{k,vn}(t) = \sum_{m=1}^M \alpha_{m,k,vn}(t) \mathbf{b}_m(\mathbf{z}_{k,vn}(t))$$

